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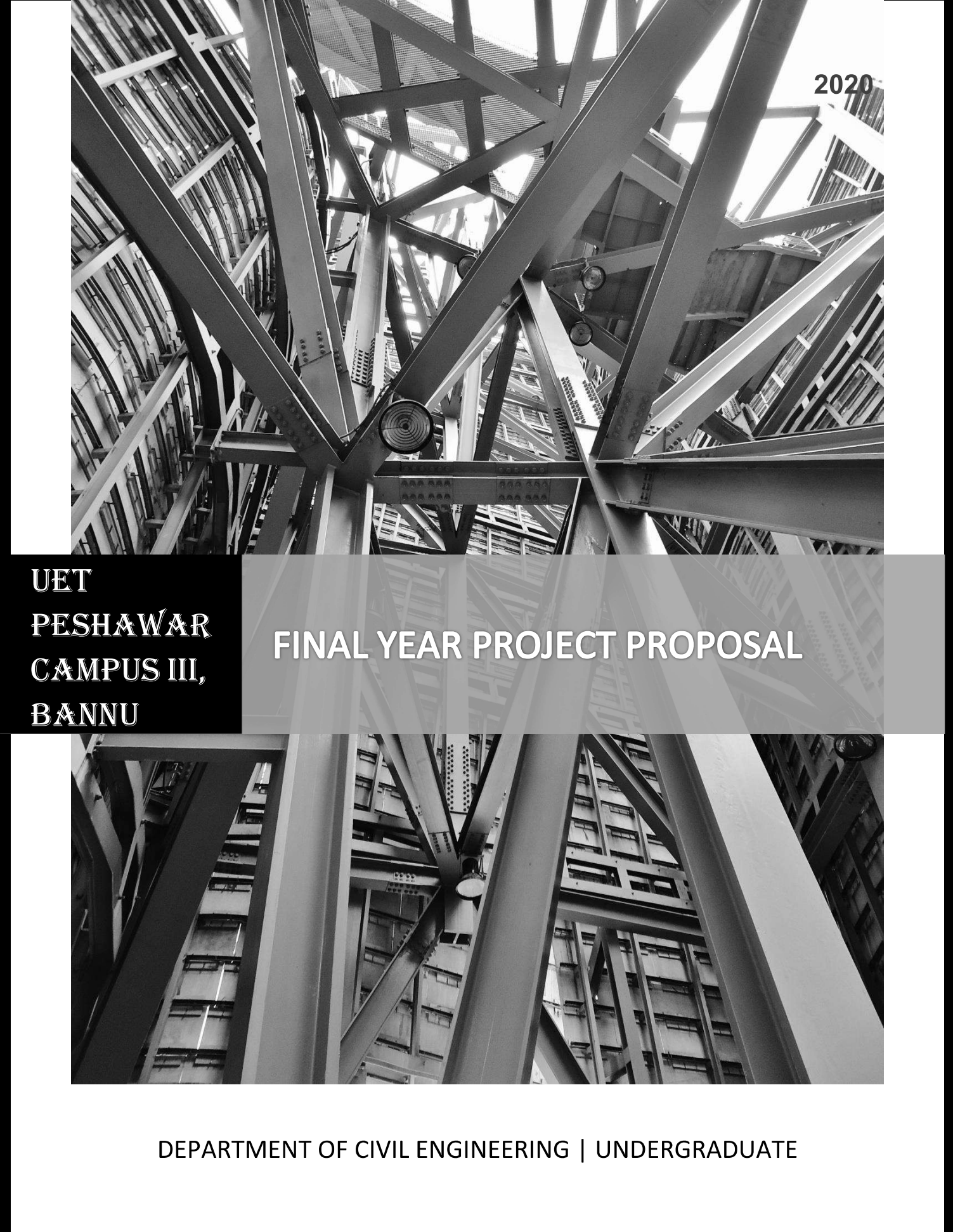


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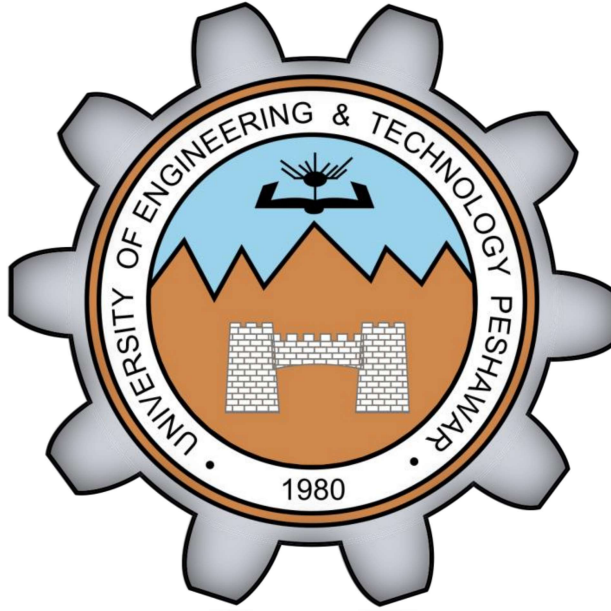


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FINAL YEAR PROJECT PROPOSAL

DEPARTMENT OF CIVIL ENGINEERING | UNDERGRADUATE



FINAL YEAR PROJECT PROPOSAL

(UNDERGRADUATE)

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NUMERICAL MODELING & EXPERIMENTAL VALIDATION OF THE BEHAVIOUR OF REINFORCED CONCRETE SANDWICHED PANELS (RCSPs) UNDER BLAST LOAD

1. Introduction

The use of bombs to attack city centers has been a feature of campaigns by terrorist organizations around the world. A bomb explosion within or immediately nearby a building can cause catastrophic damage on the building's external and internal structural frames, collapsing of walls, blowing out of large expanses of windows, and shutting down of critical life-safety systems. Loss of life and injuries to occupants can result from many causes, including direct blast-effects, structural collapse, debris impact, fire and smoke. The indirect effects can combine to inhibit or prevent timely evacuation, thereby contributing to additional casualties. In addition, major catastrophes resulting from gas-chemical explosions result in large dynamic loads, greater than the original design loads, of many structures. Due to the threat from such extreme loading conditions, efforts have been made during the past three decades to develop methods of structural analysis and design to resist blast loads (Ngo et al., 2007). It is one of the biggest challenges for structural engineers to design structures which have better impact-absorbing properties.

Much research has been carried out on dynamic response of the reinforced concrete walls subjected to explosion load. Sandwich structures with porous cores, such as metal foams and honeycombs, exhibit superior weight-to-stiffness and strength properties in explosion-resistance structural applications (Abbas et al., 2019).

While the demand is increasing, researchers are looking for a combination of such construction materials which are economical, impact-absorbent and also durable. One of the choices, presently, is expanded polystyrene (EPS) foam which can be sandwiched between two sheets of concrete called wythes (Benayoune et al., 2004) and the assembly can be reinforced with steel.

1.1 Expanded Polystyrene Foam

Expanded Polystyrene (EPS) is a white foam plastic material produced from solid beads of polystyrene. It is composed of:

- 1) **Styrene** – Which forms the cellular structure
- 2) **Pentane** – Which is used as a blowing agent

EPS is a lightweight, rigid, closed-cell insulation. It is available in several compressive strengths to withstand load and backfill forces. This closed-cell structure provides minimal water absorption and low vapour permanence. EPS is 98% air and it is 100% recyclable.



Figure 1: Preparatory stages of EPS Foam

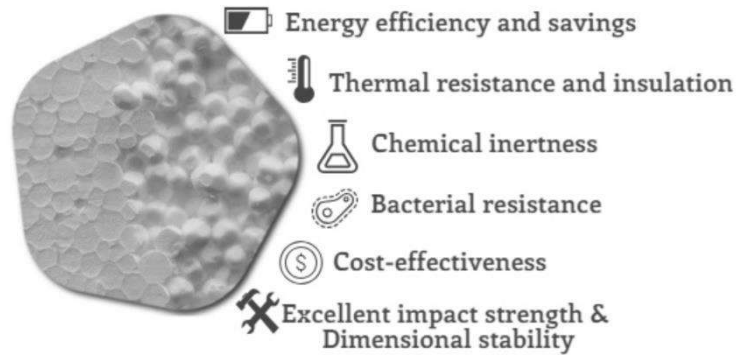


Figure 2: Advantages of EPS Foam

1.2 Reinforced Concrete Sandwiched Panel

An RCSP is composed of an EPS foam core surrounded by spray-on reinforced concrete skins (like ferrocement overlay) on both sides. A schematic of a typical RCSP is shown in Fig. 3. Research studies conducted by (Adil, 2010) have shown that RCSP has satisfactory performance under different loads i.e. shear, axial and flexure and possesses potential to be used as a possible construction technique to resist shocks (blast loads).

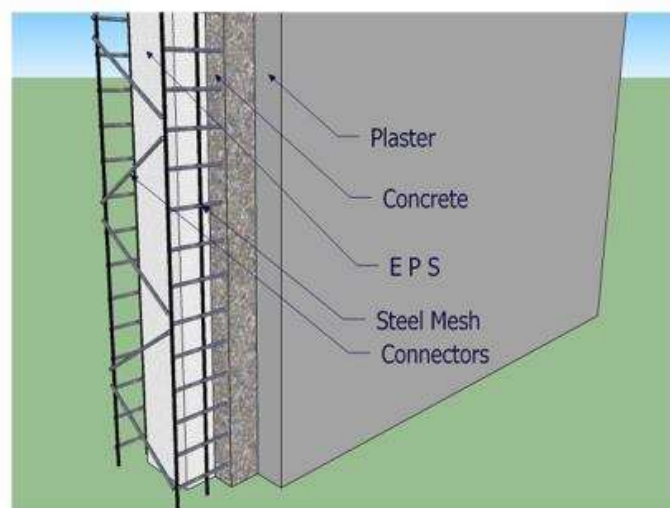


Figure 3: Reinforced concrete sandwiched panel

2. Problem Statement

Terrorist attacks are increasing day by day. The need for structures that can withstand sufficient explosive loads has been felt decades ago. One of the solutions to this problem is to build sandwiched panels but before anything is properly implemented, a lot of research must be carried out. The research on this issue includes a large number of experiments consisting of, but not limited to, making of prototypes, application of blast loads, dynamic analysis and recording of the data related to damage intensities and scaled distances. Obviously, such a research would be time-consuming and costly.

An alternative to this is to do a small scale practical work and then use a non-linear dynamic analysis software package for modeling. After a given case is simulated, it can further be extended to get more analysis results and refine the conclusions of the practical work.

3. Objectives

Objectives of the study are:

- A. To develop a numerical model for reinforced concrete sandwiched panel.
- B. To experimentally validate the behavior of reinforced concrete sandwiched panel subjected to blast load.
- C. To refine, through numerical modeling, different relationships among some important parameters for RCSP's under explosions beyond the linear range obtained from experimental work.

4. Literature Review

Efforts have been made during the past three decades to develop methods of structural analysis and design to resist blast loads (Ngo et al., 2007). Ngo et al. studied the effects of explosion on structures and concluded that requirements on ductility levels help improve the building performance under severe load conditions.

Numerical simulations and corresponding experiments of blast loading were investigated by Björn Zakrisson. The effect of the blast depended on the shape of the explosive and the state of the explosive confinement. This made it uncertain to use general analytical loading functions to describe the pressure wave in space and time in numerical models. Instead, an approach based on continuum mechanics to describe the blast load had been shown efficient, even though the choice of mesh size is critical. Also, the quality of the experimental data influenced the outcome of a validation of numerical results. However, the main conclusion from his work was that numerical simulations could be described within 3.2-5.4 % regarding air blast loading in the near-field (Zakrisson, 2010).

Several techniques have been tried by investigators to examine feasible methods for strengthening masonry walls in order to enhance their resistance to blast load. One of the most popular methods of retrofitting unreinforced masonry (URM) walls is the application of fiber reinforced polymers (FRP) to its surface. In a study the numerical model showed good correlation with experiments. The analysis was able to predict the blast pressures, damage pattern and de-bonding failure of glass fiber reinforced polymer (GFRP) sheets with reasonable accuracy. Thus it was suggested that the numerical model could be employed to predict the blast response of FRP-strengthened URM infill walls (Alsayed et al., 2016).

The *Universiti Putra Malaysia*, has undertaken extensive experimental and theoretical investigations to develop a load bearing system using Precast Concrete Sandwich Panel (PCSP). A description and the numerous advantages of Industrialized Building System (IBS) over the traditional building construction systems were presented and it was concluded that is a viable and economical industrialized building system (Benayoune et al., 2004). Also, Ye et al. presented a simplified model and experiment to validate the reliability of theoretical analysis and calculation of the dynamic response of PCSPs under blast loads. It was found that the PCSP has weaker blast resistance with respect to the RC slab. Therefore, the theoretical model can be used to conveniently predict the dynamic response of PCSPs, and the PCSP in precast structure is more vulnerable to the blast loads (Ye et al., 2021). The software used in this study was LS-DYNA.

Polymeric foams are widely used in industry and in domestic applications and are made of a skeleton of approximately regular open or closed cells, which are the basic unit of these materials. They have a high energy absorption capacity, are useful for shock applications, acoustic and thermal insulating problems. For these reasons, they are used in aircraft and automotive industry, buildings and packaging. Combining good mechanical properties with low density, rigid polymer foams can also be used as structural materials. Polymeric foam constitutive behavior is extremely complex on the micro-structural scale. Cellular buckling under compression initiates a long stress plateau. Further compression causes stress build up due to foam consolidation. A rate-independent elasto-plastic foam constitutive model, that features a single-surface yield criterion, had been developed. A non-associated plastic flow law and the relative density dependence showed reasonable prediction for the responses of rigid polymeric foams under monotonic loading conditions. The proposed polymeric foam model was tested with a typical foam problem and has performed adequately (Machado et al., 2011).

Oucif et al. studied cyclic and monotonic behavior of strengthened and un-strengthened square reinforced concrete columns. Two columns were taken. The first column was strengthened by glass fiber reinforced polymer and steel bars subjected to lateral cyclic loading, and the second was un-strengthened and subjected to lateral monotonic loading. The experimental behavior of RC columns was predicted numerically by means of the finite element method done in ATENA. It was found that the numerical models applied on the un-strengthened and strengthened columns reflect the behavior of the experimentally tested ones in terms of the load capacity and the maximal displacement. On the other hand, the numerical model of the

column strengthened with GFRP underestimate the load capacity compared to the experimental test which might be due to the assumption of perfect bond in the interfaces concrete/GFRP and concrete-steel-bars, and the neglecting of the FRP creep for the short term cyclic loading. The comparison of the behavior of the column strengthened with glass fiber reinforced polymer and steel bars with the un-strengthened reference column under lateral cyclic loading was also performed. It was found that the load carrying capacity and the ductility of the strengthened column are larger than the load carrying capacity and the ductility of the un-strengthened reference column, which confirms the contribution of the strengthening method to the improvement of the strength and stiffness of the column (Oucif et al., 2017).

An attempt of adaptation on one of the available numerical material models for cementitious materials for modeling geopolymers structures was made in 2018. Numerical analyses on the basis of previously performed tests were executed using ATENA. The results were presented in comparison with previous numerical analyses in Mafem3D and ABAQUS and it was confirmed that ATENA gives pretty fine result (Paszek & Krystek, 2018).

An experimental investigation was carried out to understand the behavior of reinforced concrete sandwiched panels (RCSPs) under blast load (Abbas et al., 2019). The fragility curve interpreted the relationship between damage intensity with charge size and scaled of distance. On the basis of results, it was concluded that RCSP panel had greater capability of absorbing and dissipating energy generated by blast, and stable against fragmentation as compared to other conventional building systems used. Based on the assessment, RCSPs walls/panels were strongly recommended for protective barrier installed in the vicinity of critical infrastructure against blast loading.

In 2002, Cervenka et al. described three nonlinear material models available in ATENA: crack band model based on fracture energy, fracture-plastic model with non-associated plasticity and micro-plane material model. It was stated that nonlinear simulation using these advanced constitutive models could be efficiently used to support and extend experimental investigations and to predict behavior of structures and structural details (Cervenka et al., 2002). Conclusions of this study were that the nonlinear finite element package ATENA was based on advanced constitutive models. Crack band approach had employed for tensile and compressive softening avoids the finite element mesh sensitivity of solution. And that ATENA was able to predict the behavior of steel reinforcement as well as steel anchors in concrete structures consistently. It could be effectively used to support and extend experimental investigations for innovative solutions in the field of connections between steel and concrete.

The effects of different TNT masses on both the Aramid fiber reinforced plastic AFRP sheets strengthened and non-strengthened RC slabs were studied by Kong et al. Comparison of the maximum deflection under the same TNT mass with that of the non-strengthened slabs showed that the deformations of the strengthened slabs all had a certain degree of reduction.

However, the reduction extent of the deflection of the slab under large TNT mass was less than that under moderate TNT mass (Kong et al., 2018).

Comparative analysis in nonlinear model of pre-stressed concrete beam was done by Xingjian Wang. He, for his study, took the data from an experimental work done in Austria in 2013 in the University of Natural Resources and Life Sciences and the associated Partner Laboratories. One of the main purposes of his work was to investigate the time effort required and the additional knowledge gained by modeling techniques of increasing accuracy, and based on different software packages. Wang compared the numerical modeling done in ATENA and ABAQUS. On investigation, he found that ATENA was effective in calculating the results even with a poor mesh. But ATENA spent much more time in analyzing a certain model than ABAQUS, while ABAQUS appeared more difficult for new users (Wang, 2019).

In a research, a blastworthy aluminum foam sandwich (AFS) structure that consisted of an occupant side plate (OSP), a struck side plate (SSP), and an aluminum foam (Al-foam) core were numerically and experimentally subjected to blast-fragmented loading. The explosion with high-pressure shock waves was produced by steel-covered TNT, creating a synergistic blast and fragment loading. The interaction between the blast-fragment loading and the AFS created a unique perforation pattern due to Monroe's effect. The measured blastworthiness characteristics included structural integrity, acceleration, and reaction force. A numerical modeling strategy to analyze the blastworthiness performance of the AFS structure was developed to capture the dynamic responses and the damage mechanism. The numerical model gave a good agreement with the test data (Pratomo et al., 2020)

5. Proposed Methodology

The final year project is planned to be based on the work in (Abbas et al., 2019). Given the reasons mentioned in the problem statement, numerical modeling will be done by finite element method. To accomplish objectives of the study, the following methodology will be adopted:

5.1 Numerical Modeling

After the selection of appropriate software package, the practical work will be modeled numerically. This phase will be completed through the following divisions:

5.1.1 Material Modeling

Constituents of the material model will be:

- 1) Concrete
- 2) Galvanized Iron (Steel)
- 3) Expanded Polystyrene Foam

These material properties will be defined via the specifications given in the following tables:

Table 1: Compressive strength of cylinder samples taken from concrete mix (1:2:1)

S/N	Strength (MPa)	Average Strength (MPa)
1	16.34	15
2	11.94	
3	16.34	

Table 2: Compressive strength of cube samples taken from plaster mix (1:3)

S/N	Strength (MPa)	Average Strength (MPa)
1	5.85	6.98
2	8.5	
3	6.61	

Table 3: Tensile strength of Galvanized Iron (Steel)

S/N	Nominal Diameter (mm)	Yield Strength (MPa)	Ultimate Strength (MPa)
1	2	184	272
2	2	200	232
3	2	192	264
Average	-	192	264.48

Table 4: Properties of expanded polystyrene (EPS)

S/N	Description	MPa
1	Modulus of Elasticity	0.35772
2	Shear Modulus	0.17886
3	Tensile Strength	0.01595
4	Compression	0.00736
5	Flexural Strength	0.01371
6	Shear Strength	0.01647

5.1.2 Dynamic Link Library (Conditional)

In a worst case scenario, when there is no choice left to simulate EPS in pre-existing properties, a direct link library (DLL) will be generated using computer programming like C++, FORTRAN or Pascal. Then the desired property will be implemented in modeling.

5.1.3 Static Analysis

Verification of material model will be done through static analysis. For this purpose, results of some monotonic tests on a similar panel would be taken. The model taken here may be different from that under study. Static analysis will be performed and the simulated results will be compared with the test results. The need for this step is that it would explicitly check the material definition.

5.1.4 Geometric Modeling

The prototype consists of four RCSPs arranged in a rectangular fashion. They are in pairs of two different thicknesses. None of the panels is connected to another. The foundation details show a fixed support between each panel and ground. Hence, they are cantilevered. Geometric model configuration is shown in the following figure:

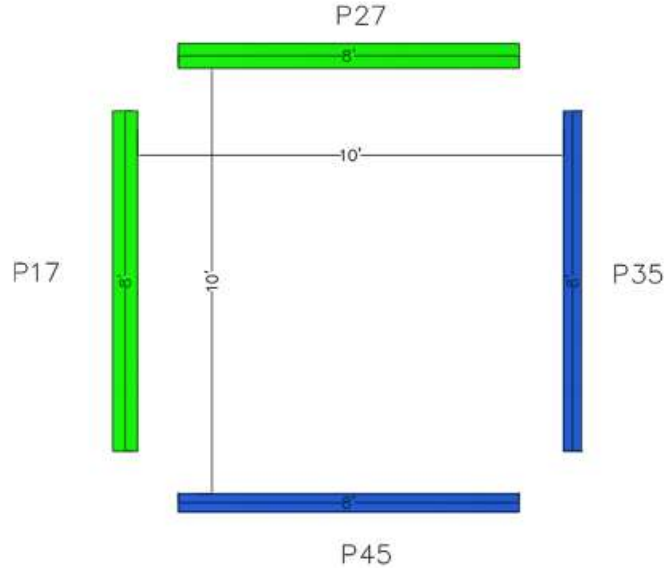


Figure 4: RCSPs Layout

Each panel is 2.4384m long and 3.048m high. Further geometry is detailed in the following table:

Table 5: RCSPs dimensional details

Panel ID	Plaster Layer Top & Bottom (mm)	Concrete Layer Top & Bottom (mm)	Polystyrene Layer Middle (mm)	Overall Thickness (mm)
P17	25.4	50.8	101.6	177.8
P27	25.4	50.8	101.6	177.8
P35	25.4	50.8	50.8	127
P45	25.4	50.8	50.8	127

5.1.5 Blast Load Application

Explosive used in the experiment was trinitro toluene (TNT), which is the most widely used material for researches related to blasts. Blast load will be simulated dynamically. Details of blasts are shown in the table below:

Table 6: Progressive blast load tests details

Blast No.	Quantity (kg)	Distance (m)	Location (Relative to enclosure)
1	0.5	0.91 from thick panel	Outside
2	1	1.1 from thick & thin panel	Inside
3	1.5	0.91 from thick & thin panel	Inside
4	2	1.2 from thick panel	Outside
5	2.5	0.91 from thick panel	Outside
6	3	0.91 from thick panel	Outside
7	3.5	0.91 from thin panel	Outside
8	4	3 from thin panel	Outside
9	4.5	1.5 from thin panel	Outside
10	5	In contact with thin panel	Outside

Following is an illustration of the locations of blasts:

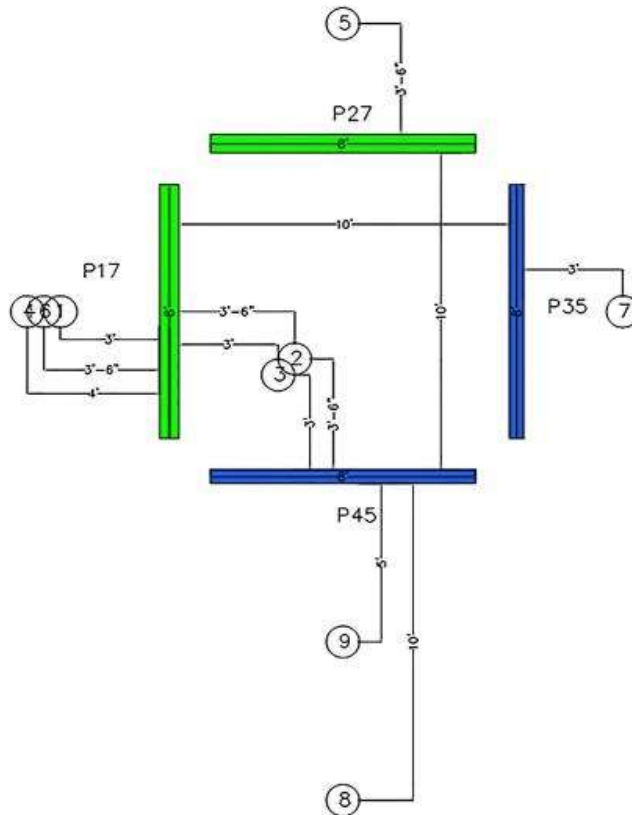


Figure 5: Progressive blast load testing protocols, locations and distance from

5.1.6 Dynamic Analysis

After the experimentation is modeled, explicit dynamic analysis will be performed using different mesh configurations and selecting the most suitable one. All the inputs that are required for the experimental validation will be completed and revised at this phase.

5.2 Laboratory Testing (Conditional)

In case any extra data is required that is not possible to be drawn from the documented practical work, laboratory tests will be performed on samples of materials under study.

5.3 Experimental Validation

The results from the dynamic analysis will be collected, relations among different parameters will be drawn and the numerical results will be compared with the experimental work. This phase will validate the practical research work and the model will be ready for the refinement.

5.4 Adjunct

Modeling and experimental validation of will be completed at this stage and further extensions will be made. These extensions will be based upon variation of load magnitude (practically TNT weight), change in the location of blast and alteration of geometric model by re-dimensioning it as a whole as well as its components (steel, concrete & EPS foam) etc.

5.5 Results & Recommendations

Results from the adjunct work will noted and the relationships & trends etc. will be refined. Collected data will be processed and organized. Necessary figures and graphs will be obtained for the final report and project thesis. Recommendations for future research will be collocated.

6. Schedule

The work is planned to be completed before the midterm examination of spring semester 2021. Meeting with the project supervisor will be attended each week. The work is segmented and each segment has a time span for its completion as depicted in the following table:

Table 7: Project Timeframe

S/N	Month	November	December	January	February	March	April
	Activity						
1	Finalization of Software						
2	Literature Review						
3	Numerical Modeling						
4	Laboratory Testing						
5	Experimental Validation						
6	Adjunct						
7	Data Processing						

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